Timothy Schaumlöffel

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AI Research Scientist

Summary

I am a research scientist specializing in evaluation and mechanistic interpretability of multimodal foundation models. My work focuses on analyzing internal representations and emergent behaviors in (multimodal) models through causal probing, representation analysis, and large-scale experimentation. I develop frameworks for systematic model diagnostics and cognitively inspired approaches to self-supervised representation learning, with publications at CVPR, NeurIPS, and ICLR.

Core Competencies: Mechanistic Interpretability | Large–Scale Experimentation | Representation Learning | Multimodal Reasoning | Model Evaluation & Diagnostics | Deep Learning | Self-Supervised Learning

Looking For: Seeking a research internship in mechanistic interpretability, multimodal reasoning, or foundation model evaluation at an industry research lab.

Selected Publications

Vision & Multimodal Models

Preprint **T. Schaumlöffel***, M.G. Vilas*, G. Roig. Contextual Inference from Single Objects in Vision-Language Models.

CVPR'26 (Highlight) **T. Schaumlöffel**, M.G. Vilas, G. Roig. Mechanisms of Object Localization in Vision–Language Models.

NeurIPS'25 **T. Schaumlöffel**, M.G. Vilas, G. Roig. Uncovering Object Localization Mechanisms in VLMs. *Mechanistic Interpretability Workshop.*

NeurIPS'23 M.G. Vilas, **T. Schaumlöffel**, G. Roig. Analyzing Vision Transformers for Image Classification in Class Embedding Space.

AASP'23 **T. Schaumlöffel**, M.G. Vilas, G. Roig. PEACS: Prefix Encoding for Auditory Caption Synthesis. *Workshop on Detection and Classification of Acoustic Scenes and Events*

Bio-Inspired Representation Learning

ICLR'26 **T. Schaumlöffel***, A. Aubret*, G. Roig, J. Triesch. Temporal Slowness in Central Vision Drives Semantic Object Learning.

ICDL'24 **T. Schaumlöffel***, A. Aubret*, G. Roig, J. Triesch. Learning Object Semantic Similarity with Self-Supervision.

ICDL'23 **T. Schaumlöffel***, A. Aubret*, G. Roig, J. Triesch. Caregiver Talk Shapes Toddler Vision: A Computational Study of Dyadic Play.

Neuro-AI & Model Alignment

Frontiers in Neuroinformatics D. Bersch, M.G. Vilas, S. Saba-Sadiya, **T. Schaumlöffel**, K. Dwivedi, C. Sartzetaki, R.M. Cichy, G. Roig. Net2Brain: A Toolbox to Compare Artificial Vision Models with Human Brain Responses

($*$ denotes shared contribution)

Research Experience

2023 – present **Doctoral Researcher**, *CVAI Lab, Goethe University Frankfurt*, Germany.

- Designed and executed large-scale causal probing and ablation studies on vision–language models and Vision Transformers (e.g., CLIP, LLaVA, InternVL), investigating internal representations and emergent behaviors; resulted in publications at CVPR'26 and NeurIPS'25.
- Developed analytical frameworks for examining classification and representation dynamics in Vision Transformers (NeurIPS'23).
- Built and maintain `net2brain`, a widely adopted open-source analysis framework for comparing vision model representations with neural data, adopted by 30+ research institutions. (Frontiers)

Expected 2026 **Visiting Researcher**, *Indian Institute of Technology Bombay*, India.

- Visiting researcher within the DAAD–DST Indo-German research exchange on adaptive deep neural networks under distribution shift.
- Collaborated on continual learning and evaluation strategies for multimodal foundation models.

2023 – 2025 **Research Collaboration**, *Frankfurt Institute for Advanced Studies*, Germany.

- Developed self-supervised learning methods that extract structured representations from temporal data streams. (ICLR'26, ICDL'24, ICDL'23).
- Contributed to interdisciplinary research connecting neuroscience-inspired learning principles with self-supervised vision model design and evaluation.

2024 – 2026 **Research Collaboration**, *Institute of Psychology, Goethe University Frankfurt*, Germany.

- Designed multimodal analysis pipelines combining egocentric video, gaze signals, and audio to study learning dynamics in naturalistic environments.
- Developed object-centric temporal analysis methods using detection and tracking to quantify attention and exploration patterns.
- Connected cognitively inspired learning signals with evaluation frameworks for modern vision–language models.

Technical Skills

Foundation Models	LLMs, Vision Transformers (ViTs), Vision–Language Models (VLMs), multimodal reasoning systems
Training & Scaling	Self-supervised and contrastive learning; instruction tuning and fine-tuning; distributed training (DDP, FSDP); multi-node/multi-GPU experimentation; mixed-precision and memory-efficient training
Evaluation	Mechanistic interpretability; causal ablations; representation analysis (CKA, RSA); probing and model diagnostics; robustness and generalization evaluation
Development	Python, PyTorch, Hugging Face Transformers, vLLM, Linux, Git, CI/CD, software testing, reproducible experiment design
Infrastructure	HPC environments; large-scale experiment pipelines; dataset processing and benchmarking workflows
Open Source	Core contributor to `net2brain`

Education

2023–present **PhD Candidate, Computer Science (AI & Multimodal Learning)**, *Goethe University Frankfurt*, Germany.

- Thesis: Inner Interpretability of AI Models – mechanistic analysis of multimodal foundation models and bio-inspired learning systems
- Supervised by Prof. Gemma Roig

2021–2023 **Master of Science in Computer Science with Artificial Intelligence**, *Goethe University Frankfurt*, Germany.

- Grade: 1.0 with honors (ranked 1st in class), Thesis: 1.0
- Awarded a merit-based academic excellence prize

2017–2021 **Bachelor of Science in Computer Science**, *Goethe University Frankfurt*, Germany.
- Grade: 1.4, Thesis: 1.3
- Minor: Physics

Fellowships & Awards

2025–2027 DAAD–DST Indo-German Research Fellowship (Competitive international research grant) – *Adaptive Deep Neural Networks under Continually Changing Environments*

2023 Best Paper Award ICDL'23

2023 Honours Masters Degree, Goethe University Frankfurt

Service Reviewer in NeurIPS, ICLR, CVPR, WACV, IEEE ICLD

Teaching & Mentorship

2022–2026 **Instructor & Project Mentor**, *Deep Learning for Computer Vision*.
Led and delivered full course lectures; designed coursework and exams; supervised student projects

2018–2021 **Teaching Assistant**, *Core Computer Science*.
Tutored and mentored students in theoretical computer science, databases, and hardware design

2024–2026 **Selected Thesis Supervision at Goethe University Frankfurt:**
- Explainability of key–value memory pairs in Vision Transformers
- Domain generalization methods for visual learning
- Loss function effects in large vision–language models
- AI-based analysis of attachment style and exploration in toddlers